

# DIABETIC RETINOPATHY SCREENING REPORT

AI-Assisted Fundus Image Analysis — with Bootstrap 95% CI

**Image:** idrid\_IDRiD\_0596.jpg  
**Date:** 2026-03-28 17:30:59  
**Model:** EfficientNet-B3 (Binary + Multiclass Cascade)  
**Bootstrap:** 2000 resamples, 95% confidence level

## SCREENING RESULT (with 95% Bootstrap CI)

### Binary Detection:

- Ensemble (5-fold+TTA): DR Detected  
P(DR) = 0.9345 95% CI [0.8711, 0.9785]
- 2-Phase (single+TTA): DR Detected  
P(DR) = 0.9960 95% CI [0.9926, 0.9983]

### Severity Grading (Direct Multiclass Model):

- Predicted Grade: 4 — Proliferative DR
- Confidence: 66.4% 95% CI [0.5251, 0.7965]
- Severity: Sight-threatening

### Per-Grade Probabilities (95% CI):

|                       |           |                    |
|-----------------------|-----------|--------------------|
| G0 (No DR)            | ): 0.0354 | [0.0224, 0.0489]   |
| G1 (Mild NPDR)        | ): 0.0562 | [0.0241, 0.0957]   |
| G2 (Moderate NPDR)    | ): 0.0753 | [0.0439, 0.1147]   |
| G3 (Severe NPDR)      | ): 0.1688 | [0.0894, 0.2589]   |
| G4 (Proliferative DR) | ): 0.6644 | [0.5251, 0.7965] ◀ |

### Clinical Recommendation:

- **URGENT referral — risk of vision loss**

*DISCLAIMER: This is an AI screening aid, not a diagnostic tool. All results must be confirmed by a qualified ophthalmologist.*

# Preprocessing Pipeline — idrid\_IDRiD\_0596.jpg

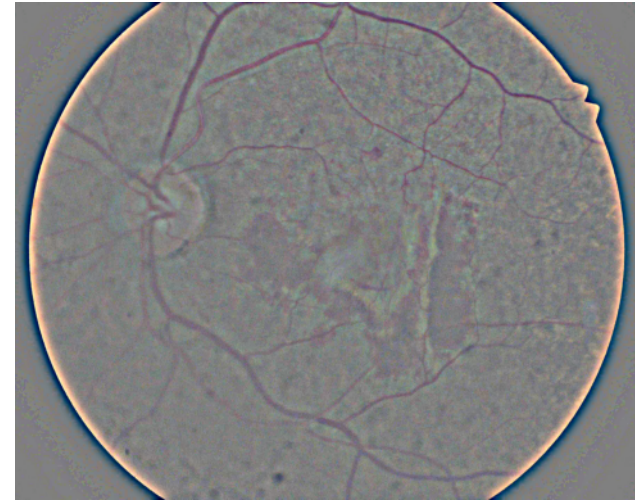
Step 1: Original



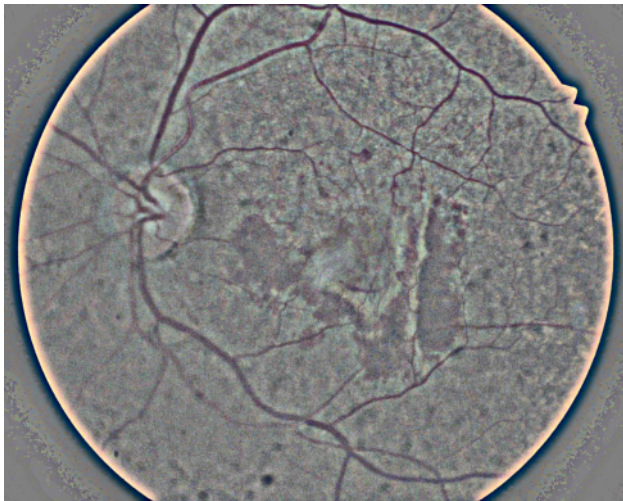
Step 2: Circle Crop



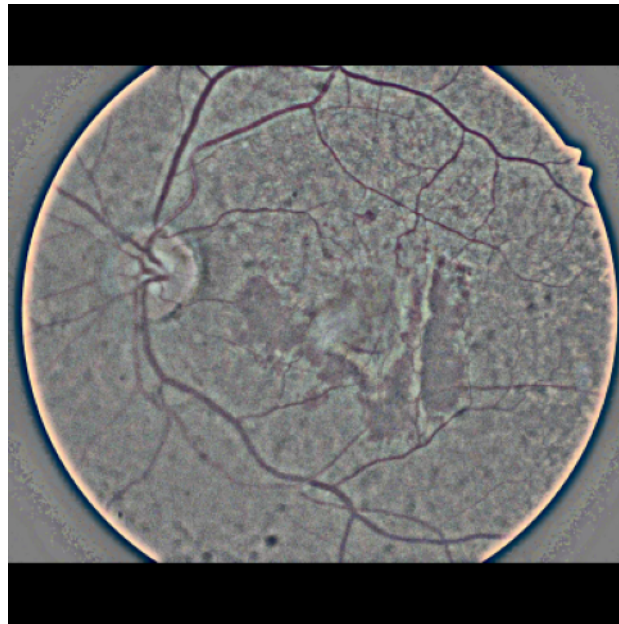
Step 3: Ben Graham



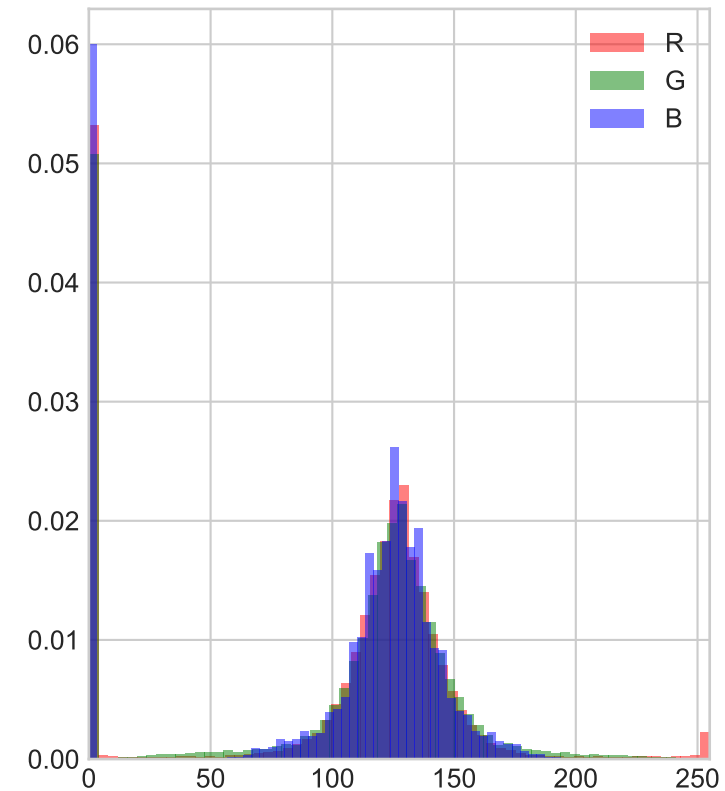
Step 4: CLAHE



Step 5: Final (384×384)

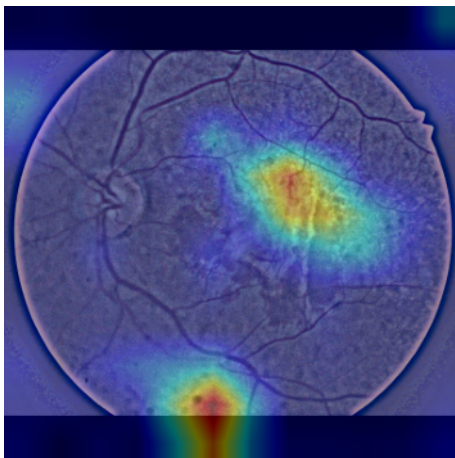


Preprocessed Histograms

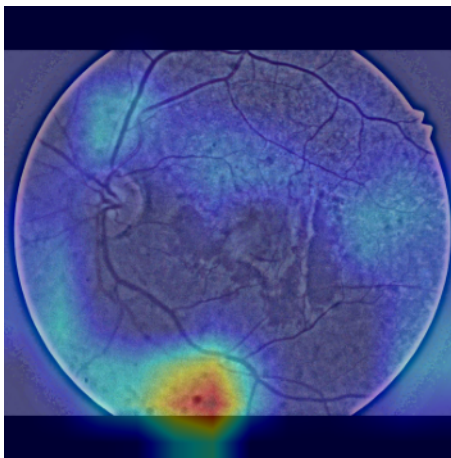


# Explainability & Predictions with 95% CI — idrid\_IDRiD\_0596.jpg

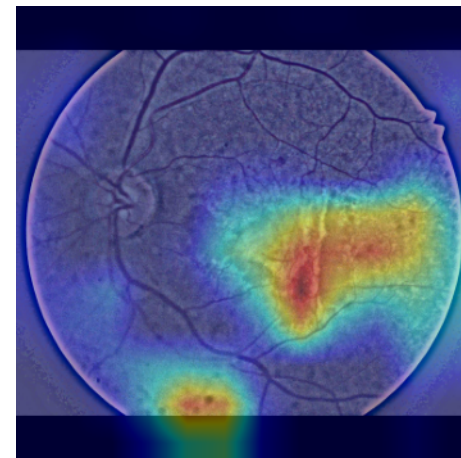
**Ensemble GradCAM**  
P(DR)=0.9345 CI[0.8711,0.9785]



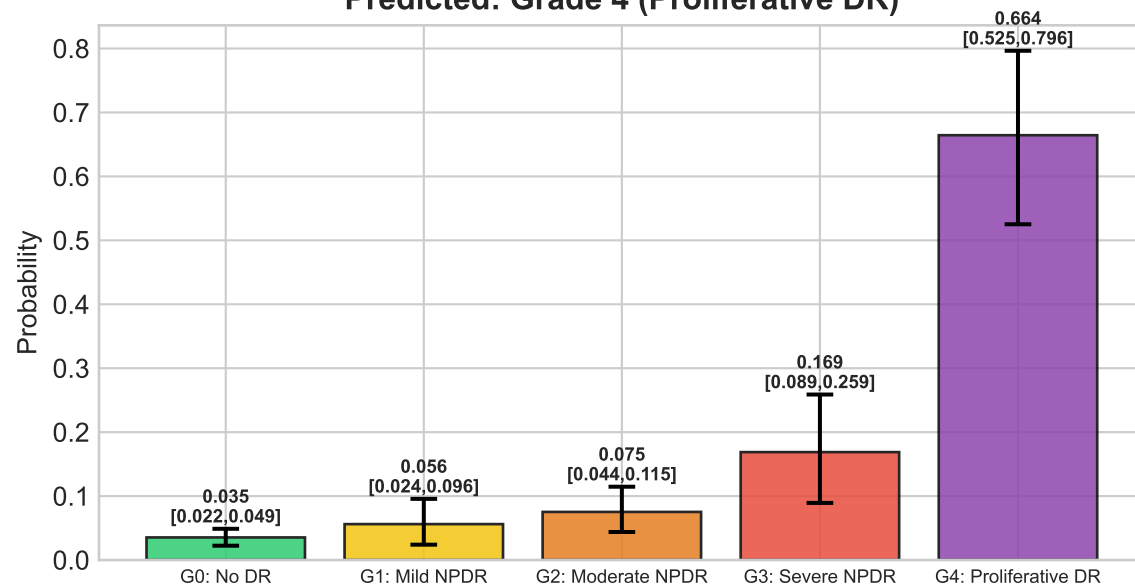
**2-Phase GradCAM**  
P(DR)=0.9960 CI[0.9926,0.9983]



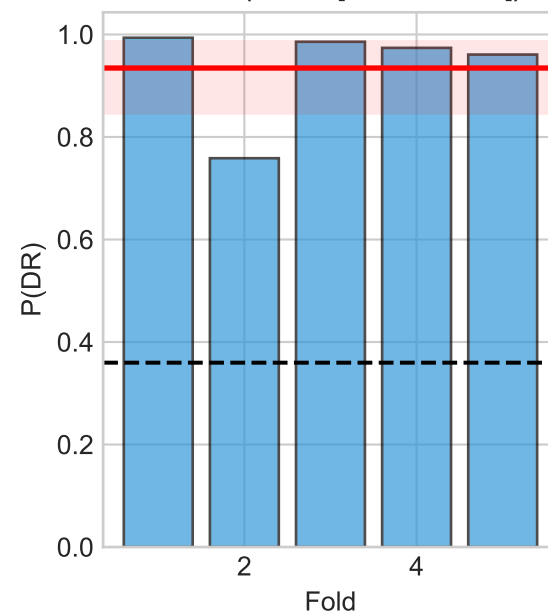
**Multiclass GradCAM (G4)**  
Conf=66.44% CI[0.525,0.796]



**5-Grade Probabilities with 95% CI (n\_boot=2000)**  
**Predicted: Grade 4 (Proliferative DR)**



**Per-Fold (95% CI [0.8460, 0.9865])**



# Model Validation Summary

## MODEL VALIDATION METRICS

Dataset: APTOS 2019 (531 test images)

Bootstrap: 2000 resamples, 95% CI

| Metric      | Ensemble+TTA | 2-Phase+TTA |
|-------------|--------------|-------------|
| AUC-ROC     | 0.9992       | 0.9992      |
| Accuracy    | 0.9906       | 0.9887      |
| Sensitivity | 0.9962       | 0.9962      |
| Specificity | 0.9852       | 0.9815      |
| F1-Score    | 0.9905       | 0.9886      |
| Cohen Kappa | 0.9812       | 0.9774      |

Multiclass Model (Direct 5-Class):

Best Validation QWK: 0.859

Architecture: EfficientNet-B3 + Dropout(0.4) + Linear(1536, 5)

Training: CrossEntropyLoss + label\_smoothing=0.1

## THIS IMAGE – PREDICTION SUMMARY

Ensemble P(DR): 0.934528 95% CI [0.871123, 0.978543]

2-Phase P(DR): 0.995971 95% CI [0.992558, 0.998269]

Multiclass Grade: 4 (Proliferative DR)

Confidence: 0.6644 95% CI [0.5251, 0.7965]

Per-Grade Probabilities (95% CI):

G0 No DR : 0.0354 [0.0224, 0.0489]

G1 Mild NPDR : 0.0562 [0.0241, 0.0957]

G2 Moderate NPDR : 0.0753 [0.0439, 0.1147]

G3 Severe NPDR : 0.1688 [0.0894, 0.2589]

G4 Proliferative DR : 0.6644 [0.5251, 0.7965]